

N-dimensional Permutational Invariant Coefficient Mapping

Phil Alcorn

June 2026

The goal is to map a set of n-dimensional coefficients $C_{a,b,c,...n}$ to a unique index i that avoids duplication due to permutations.

This means, as an example, that $C_{1,0,0}$, $C_{0,1,0}$ and $C_{0,0,1}$ must map to the same i .

There are two levels of permutation invariance. Each specific atom combination has a set of coefficients for each Chebyshev polynomial that is permutation invariant. Additionally, the set of atoms represented in a system is also permutation invariant. To ensure locality, these two sets must be combined in a way that that the sets of coefficients for a given set of atoms will remain together in memory.

This means that the coefficients are mapped in the following order (assuming 3 distinct atom types and a maximum chebyshev order of 2). Note that each element in this list represents a single coefficient- the combination of E-C subscripts only serves to indicate what that coefficient represents in the energy equation.

$$\begin{aligned} & [E_{0,0,0}C_{0,0,0}, E_{0,0,0}C_{0,0,1}, E_{0,0,0}C_{0,0,2}, E_{0,0,0}C_{0,1,1}, E_{0,0,0}C_{0,1,2}, E_{0,0,0}C_{1,1,1}, E_{0,0,0}C_{1,1,2}, \\ & E_{0,0,0}C_{1,2,2}, E_{0,0,0}C_{2,2,2}, E_{0,0,1}C_{0,0,0}, E_{0,0,1}C_{0,0,1}, E_{0,0,1}C_{0,0,2}, E_{0,0,1}C_{0,1,1}, E_{0,0,1}C_{0,1,2}, \\ & E_{0,0,1}C_{1,1,1}, E_{0,0,1}C_{1,1,2}, E_{0,0,1}C_{1,2,2}, E_{0,0,1}C_{2,2,2}, E_{0,0,2}C_{0,0,0}, \dots, E_{m,m,m}C_{n,n,n}] \end{aligned}$$

with m representing the number of distinct atoms and n representing the maximum order of polynomials. The number of E -indices represents the number of atoms in a given system, and the number of c -indices represents the number of 2-body pairs in a system. The ordering of each set of subscripts is permutation invariant.

The key concept to note is that this ordering as a combination of two sets of permutation invariant indices is different than the ordering of a single set of six permutation invariant indices, which would look like the following:

$$\begin{aligned} & [C_{0,0,0,0,0,0}, C_{0,0,0,0,0,1}, C_{0,0,0,0,0,2}, C_{0,0,0,0,1,1}, C_{0,0,0,0,1,2}, C_{0,0,0,0,1,2}, C_{0,0,0,0,2,2}, \\ & C_{0,0,0,1,1,1}, C_{0,0,0,1,1,2}, C_{0,0,0,1,2,2}, C_{0,0,0,2,2,2}, C_{0,0,1,1,1,1}, C_{0,0,1,1,1,2}, C_{0,0,1,1,2,2}, \\ & C_{0,0,1,2,2,2}, C_{0,0,2,2,2,2}, C_{0,1,1,1,1,1}, \dots, C_{n,n,n,n,n,n}] \end{aligned}$$

Due to this ordering, the calculation of the index must be split into multiple sections. Since the coefficients are ordering in a single dimensional linear list C , for a given coefficient $C[\alpha]$ we can find its index α using

$$\alpha = N_c i + j$$

Where N_c is the number of coefficients per element combination (or, the length of the list $[c_{0,0,0}, \dots, c_{n,n,n}]$), i is the element pair offset (or, the location of $E_{system}c_{0,0,0}$, and j is the specific coefficient offset, $E_{system}c_{desired} - E_{system}c_{0,0,0}$.

N_c is given by the standard stars and bars method, which gives the number of ways to choose a non-decreasing sequence of length k from n distinct values (i.e. a multiset of size k) is given by:

$$N_c = \binom{n+k-1}{k} = \binom{(o+1) + \binom{b}{2} - 1}{\binom{b}{2}} = \binom{o + \binom{b}{2}}{\binom{b}{2}}$$

where o is the maximum Chebyshev order (so there are $o+1$ possible values: $0, 1, \dots, o$), and $\binom{b}{2}$ is the number of 2-body pairs.

To find i and j , the Coefficient indices and Element indices must be sorted and transformed into a strictly increasing set. This is because the combinadic representation of a set of indices can only guarantee a unique integer for each set iff the set indices in the set are strictly increasing. For an index $[i_1, i_2, \dots, i_d]$ sorted in non-decreasing order, we convert to a strictly increasing index by adding $[0, 1, 2, \dots, d-1]$:

$$[i_1, i_2, \dots, i_d] \mapsto [i_1, i_2 + 1, i_3 + 2, \dots, i_d + (d-1)] = [c_1, c_2, \dots, c_d]$$

The original index value is then recovered from the transformed index $[c_1, c_2, \dots, c_d]$ via the combinadic representation:

$$N = \binom{c_1}{1} + \binom{c_2}{2} + \dots + \binom{c_d}{d} = \sum_{k=1}^d \binom{c_k}{k}$$

where $c_1 < c_2 < \dots < c_d$.

Here, N is computed twice: once for the E indices with d equal to the number of atoms in the system, and c_k coming from the transformed set of element indices (giving i), and once for the C indices with d equal to the number of 2-body pairs, and c_k coming from the transformed set of coefficient indices (giving j).